

Merge

<u>A</u>	<u>B</u>	<u>C</u>
$i \rightarrow 2$	$5 \leftarrow j$	$2 \leftarrow k$
8	9	5
15	12	8
$i \rightarrow 18$	17	9
	19 $\leftarrow j$	12
	25	15
	30	17
		18 $\leftarrow k$

```
Algorithm Merge(A, B, m, n)
{
    i = 1; j = 1; k = 1;
    while(i <= m & j <= n)
    {
        if(A[i] < B[j])
            C[k++] = A[i++];
        else
            C[k++] = B[j++];
    }
    for(; i <= m; i++)
        C[k++] = A[i];
    for(; j <= n; j++)
        C[k++] = B[j];
}
```



Merge

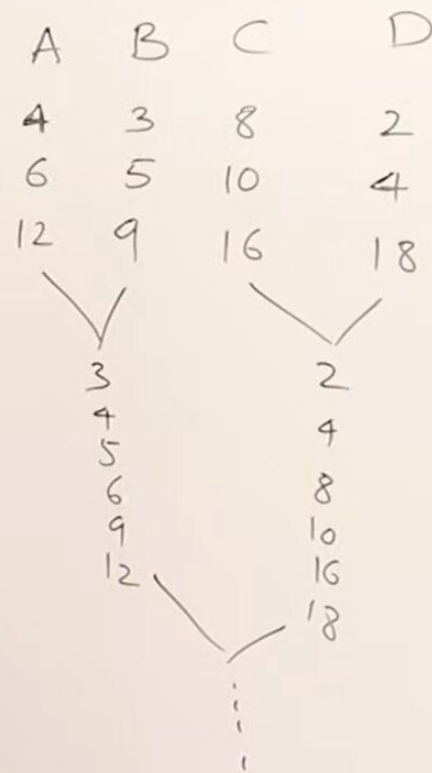
A	B	C	D
4	3	8	2
6	5	10	4
12	9	16	18

$\frac{L}{2}$
2
3
4
4

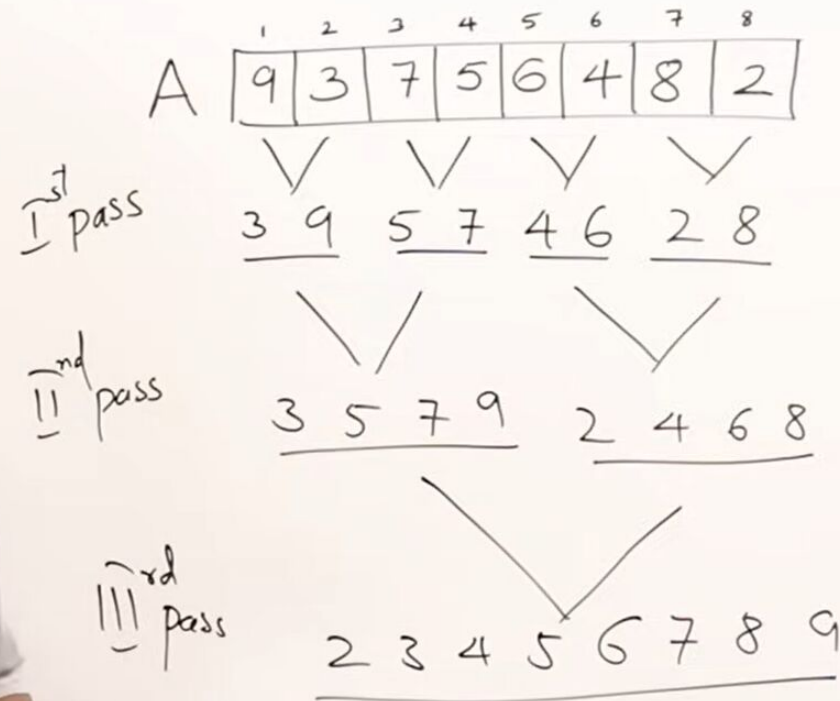
4-way Merging
M-way Merging
2-way "



Merge



2-way Merge Sort



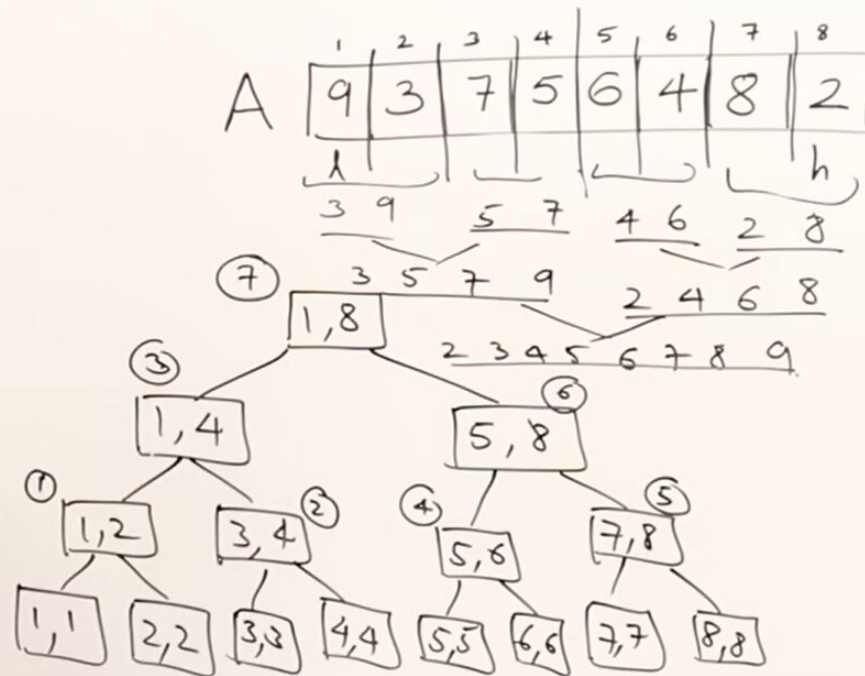
Merge Sort

	1	2	3	4	5	6	7	8
A	9	3	7	5	6	4	8	2
	l			↑ mid				h

```
Algorithm MergeSort(l, h)
{
    if (l < h)
    {
        mid = (l + h) / 2;
        MergeSort(l, mid);
        MergeSort(mid + 1, h);
        Merge(l, mid, h);
    }
}
```



Merge Sort



Algorithm MergeSort(l, h)

```

{
  if (l < h)
  {
    mid = (l + h) / 2;
    MergeSort(l, mid);
    MergeSort(mid + 1, h);
    Merge(l, mid, h);
  }
}
  
```

